

pdf-dc99cf88 — formula report

344 inline formulas, 46 display equations, 0 tables, 4 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
pdf-dc99cf88_EQ0021	010	0.263 C +10	$\begin{aligned} \hat{\rho}_{\mathrm{dog}} &= \sum_{(a,b) \in A \times B} \pi(a,b, \text{dog}, \text{barks}) \text{barks} \otimes \text{barks}^* \\ &+ \sum_{(a,b) \in A \times B} \sqrt{\pi(a,b, \text{dog}, \text{barks}) \pi(a,b, \text{dog}, \text{runs})} \text{barks} \otimes \text{runs}^* + \\ &+ \sum_{(a,b) \in A \times B} \sqrt{\pi(a,b, \text{dog}, \text{runs}) \pi(a,b, \text{dog}, \text{barks})} \text{runs} \otimes \text{barks}^* + \\ &+ \sum_{(a,b) \in A \times B} \pi(a,b, \text{dog}, \text{runs}) \text{runs} \otimes \text{runs}^* \\ &= \frac{1}{5} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}. \end{aligned}$	$\hat{\rho}_{\mathrm{dog}} = \sum_{(a,b) \in A \times B} \pi(a,b, \text{dog}, \text{barks}) \text{barks} \otimes \text{barks}^* + \sum_{(a,b) \in A \times B} \sqrt{\pi(a,b, \text{dog}, \text{barks}) \pi(a,b, \text{dog}, \text{runs})} \text{barks} \otimes \text{runs}^* + \sum_{(a,b) \in A \times B} \sqrt{\pi(a,b, \text{dog}, \text{runs}) \pi(a,b, \text{dog}, \text{barks})} \text{runs} \otimes \text{barks}^* + \sum_{(a,b) \in A \times B} \pi(a,b, \text{dog}, \text{runs}) \text{runs} \otimes \text{runs}^* = \frac{1}{5} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$	$\hat{\rho}_{\mathrm{dog}} = \sum_{(a,b) \in A \times B} \pi(a,b, \text{dog}, \text{barks}) \text{barks} \otimes \text{barks}^* + \sum_{(a,b) \in A \times B} \sqrt{\pi(a,b, \text{dog}, \text{barks}) \pi(a,b, \text{dog}, \text{runs})} \text{barks} \otimes \text{runs}^* + \sum_{(a,b) \in A \times B} \sqrt{\pi(a,b, \text{dog}, \text{runs}) \pi(a,b, \text{dog}, \text{barks})} \text{runs} \otimes \text{barks}^* + \sum_{(a,b) \in A \times B} \pi(a,b, \text{dog}, \text{runs}) \text{runs} \otimes \text{runs}^* = \frac{1}{5} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$
pdf-dc99cf88_EQ0009	006	0.272 C +10	$\psi = \frac{1}{2}(\text{big} \otimes \text{mountain} + \text{tall} \otimes \text{mountain} + \text{cold} \otimes \text{winter} + \text{chilly} \otimes \text{winter})$	$\psi = \frac{1}{2}(\text{big} \otimes \text{mountain} + \text{tall} \otimes \text{mountain} + \text{cold} \otimes \text{winter} + \text{chilly} \otimes \text{winter})$	$\psi = \frac{1}{2}(\text{big} \otimes \text{mountain} + \text{tall} \otimes \text{mountain} + \text{cold} \otimes \text{winter} + \text{chilly} \otimes \text{winter})$
Conf. MathPix confidence: ≥0.9 0.5–0.9 <0.5 — no value Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
pdf-dc99cf88_E00020	010	■0.342 ●W +1	$\begin{aligned} \rho_D &= \sum_{(a,b,c) \in A \times B \times C} \pi(a,b,c, \text{barks}) \text{barks} \otimes \text{barks}^* \\ &+ \sum_{(a,b,c) \in A \times B \times C} \sqrt{\pi(a,b,c, \text{barks})\pi(a,b,c, \text{runs})} \text{barks} \otimes \text{runs}^* \\ &+ \sum_{(a,b,c) \in A \times B \times C} \sqrt{\pi(a,b,c, \text{runs})\pi(a,b,c, \text{barks})} \text{runs} \otimes \text{barks}^* \\ &+ \sum_{(a,b,c) \in A \times B \times C} \pi(a,b,c, \text{runs}) \text{runs} \otimes \text{runs}^* \\ &= \frac{1}{5} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \end{aligned}$	$\rho_D = \sum_{(a,b,c) \in A \times B \times C} \pi(a,b,c, \text{barks}) \text{barks} \otimes \text{barks}^* + \sum_{(a,b,c) \in A \times B \times C} \sqrt{\pi(a,b,c, \text{barks})\pi(a,b,c, \text{runs})} \text{barks} \otimes \text{runs}^* + \sum_{(a,b,c) \in A \times B \times C} \sqrt{\pi(a,b,c, \text{runs})\pi(a,b,c, \text{barks})} \text{runs} \otimes \text{barks}^* + \sum_{(a,b,c) \in A \times B \times C} \pi(a,b,c, \text{runs}) \text{runs} \otimes \text{runs}^* = \frac{1}{5} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$	$\rho_D = \sum_{(a,b,c) \in A \times B \times C} \pi(a,b,c, \text{barks}) \text{barks} \otimes \text{barks}^* + \sum_{(a,b,c) \in A \times B \times C} \sqrt{\pi(a,b,c, \text{barks})\pi(a,b,c, \text{runs})} \text{barks} \otimes \text{runs}^* + \sum_{(a,b,c) \in A \times B \times C} \sqrt{\pi(a,b,c, \text{runs})\pi(a,b,c, \text{barks})} \text{runs} \otimes \text{barks}^* + \sum_{(a,b,c) \in A \times B \times C} \pi(a,b,c, \text{runs}) \text{runs} \otimes \text{runs}^* = \frac{1}{5} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$
pdf-dc99cf88_E00045	016	■0.432 ●N +0	$\mathcal{L}(s,s') = \begin{cases} \pi(s' s) & \text{if } s \leq s' \\ 0 & \text{otherwise} \end{cases}$	$\mathcal{L}(s,s') = \begin{cases} \pi(s' s) & \text{if } s \leq s' \\ 0 & \text{otherwise} \end{cases}$	$\mathcal{L}(s,s') = \begin{cases} \pi(s' s) & \text{if } s \leq s' \\ 0 & \text{otherwise} \end{cases}$
pdf-dc99cf88_E00008	006	■0.526 ●C +10	$T = \{(\text{big}, \text{mountain}), (\text{tall}, \text{mountain}), (\text{cold}, \text{winter}), (\text{chilly}, \text{winter})\}.$	$T = \{(\text{big}, \text{mountain}), (\text{tall}, \text{mountain}), (\text{cold}, \text{winter}), (\text{chilly}, \text{winter})\}.$	$T = \{(\text{big}, \text{mountain}), (\text{tall}, \text{mountain}), (\text{cold}, \text{winter}), (\text{chilly}, \text{winter})\}.$
pdf-dc99cf88_E00042	015	■0.568 ●W +1	$\hat{\rho}_{\text{dog}} = \sum_{i_2} \hat{\rho}_{x_{i_2} \text{dog}} = \sum_{i_3, i_2} \hat{\rho}_{x_{i_3} x_{i_2} \text{dog}} = \cdots = \sum_{i_{N-1}, \dots, i_3, i_2} \hat{\rho}_{x_{i_{N-1}} \cdots x_{i_3} x_{i_2} \text{dog}}.$	$\hat{\rho}_{\text{dog}} = \sum_{i_2} \hat{\rho}_{x_{i_2} \text{dog}} = \sum_{i_3, i_2} \hat{\rho}_{x_{i_3} x_{i_2} \text{dog}} = \cdots = \sum_{i_{N-1}, \dots, i_3, i_2} \hat{\rho}_{x_{i_{N-1}} \cdots x_{i_3} x_{i_2} \text{dog}}.$	$\hat{\rho}_{\text{dog}} = \sum_{i_2} \hat{\rho}_{x_{i_2} \text{dog}} = \sum_{i_3, i_2} \hat{\rho}_{x_{i_3} x_{i_2} \text{dog}} = \cdots = \sum_{i_{N-1}, \dots, i_3, i_2} \hat{\rho}_{x_{i_{N-1}} \cdots x_{i_3} x_{i_2} \text{dog}}.$
pdf-dc99cf88_E00039	014	■0.613 ●W +1	$\begin{aligned} \text{I climbed} &\leq \text{I climbed the tall mountain} \\ \text{a hot summer} &\leq \text{iced tea on a hot summer day} \\ \text{dog} &\leq \text{black dog} \leq \text{small black dog} \end{aligned}$	$\begin{aligned} \text{I climbed} &\leq \text{I climbed the tall mountain} \\ \text{a hot summer} &\leq \text{iced tea on a hot summer day} \\ \text{dog} &\leq \text{black dog} \leq \text{small black dog} \end{aligned}$	$\begin{aligned} \text{I climbed} &\leq \text{I climbed the tall mountain} \\ \text{a hot summer} &\leq \text{iced tea on a hot summer day} \\ \text{dog} &\leq \text{black dog} \leq \text{small black dog} \end{aligned}$
pdf-dc99cf88_E00043	015	■0.715 ●C -3	$\text{dog} \leq \text{black dog} \rightsquigarrow \text{dog} \stackrel{\pi(\text{black} \text{dog})}{\leq} \text{black dog}$	$\text{dog} \leq \text{black dog} \rightsquigarrow \text{dog} \stackrel{\pi(\text{black} \text{dog})}{\leq} \text{black dog}$	$\text{dog} \leq \text{black dog} \rightsquigarrow \text{dog} \stackrel{\pi(\text{black} \text{dog})}{\leq} \text{black dog}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
pdf-dc99cf88_E00034	013	■0.731 ● W +2	$\begin{aligned} & \backslash begin{aligned} \backslash rho_{\mathrm{dog}} \& = \pi(\backslash text{ small black } \backslash mid \backslash operatorname{dog}) \backslash rho_{\backslash text{small black dog}} \backslash \& + \pi(\backslash text{ small white } \backslash mid \backslash operatorname{dog}) \backslash rho_{\backslash text{small white dog}} \backslash \& + \pi(\backslash text{ big black } \backslash mid \backslash operatorname{dog}) \backslash rho_{\backslash text{big black dog}} \backslash \& = \frac{1}{3} \left[\frac{1}{0} \frac{0}{0} \right] + \frac{1}{3} \left[\frac{1}{0} \frac{0}{0} \right] + \frac{1}{3} \left[\frac{0}{1} \frac{0}{0} \right] . \end{aligned}$	$\begin{aligned} \rho_{\text{dog}} &= \pi(\text{ small black } \mid \text{dog}) \rho_{\text{small black dog}} \\ &+ \pi(\text{ small white } \mid \text{dog}) \rho_{\text{small white dog}} \\ &+ \pi(\text{ big black } \mid \text{dog}) \rho_{\text{big black dog}} \\ &= \frac{1}{3} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \end{aligned}$	$\begin{aligned} \rho_{\text{dog}} &= \pi(\text{small black} \mid \text{dog}) \rho_{\text{small black dog}} \\ &+ \pi(\text{small white} \mid \text{dog}) \rho_{\text{small white dog}} \\ &+ \pi(\text{big black} \mid \text{dog}) \rho_{\text{big black dog}} \\ &= \frac{1}{3} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} . \end{aligned}$
pdf-dc99cf88_E00015	008	■0.809 ● N +0	$\hat{\rho}_{x_{i_k} \cdots x_{i_1}} := \sum_{a,b} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{i_{N-1} \cdots i_1 b} x_a x_b^* .$	$\hat{\rho}_{x_{i_k} \cdots x_{i_1}} := \sum_{a,b} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{i_{N-1} \cdots i_1 b} x_a x_b^* .$	$\hat{\rho}_{x_{i_k} \cdots x_{i_1}} := \sum_{a,b} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{i_{N-1} \cdots i_1 b} x_a x_b^* .$
pdf-dc99cf88_E00031	012	■0.903 ● N +0	$\rho_{x_{i_1}} = \sum_{i_{N-1}, \dots, i_2} \pi(x_{i_{N-1}} \cdots x_{i_2} \mid x_{i_1}) \rho_{x_{i_{N-1}} \cdots x_{i_2} x_{i_1}}$	$\rho_{x_{i_1}} = \sum_{i_{N-1}, \dots, i_2} \pi(x_{i_{N-1}} \cdots x_{i_2} \mid x_{i_1}) \rho_{x_{i_{N-1}} \cdots x_{i_2} x_{i_1}}$	$\rho_{x_{i_1}} = \sum_{i_{N-1}, \dots, i_2} \pi(x_{i_{N-1}} \cdots x_{i_2} \mid x_{i_1}) \rho_{x_{i_{N-1}} \cdots x_{i_2} x_{i_1}}$
pdf-dc99cf88_E00017	008	■0.949 ● W +0	$\begin{aligned} & \backslash begin{aligned} \backslash hat{\backslash rho}_{x_{i_k}} \backslash cdots x_{i_1} \backslash \& = \sum_{\backslash substack{i_{N-1}, \backslash ldots, i_{k+1}} \backslash \& a, b} \backslash psi_{i_{N-1} \backslash cdots i_1} a \backslash bar{\backslash psi}_{i_{N-1} \backslash cdots i_1} b \backslash x_{i_k} \backslash cdots x_{i_1} \backslash \& = \sum_{i_{N-1}, \dots, i_{k+1}} \left(\sum_{a,b} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{i_{N-1} \cdots i_1 b} x_a x_b^* \right) \\ & = \sum_{i_{k+1}} \hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}} . \end{aligned}$	$\begin{aligned} \hat{\rho}_{x_{i_k} \cdots x_{i_1}} &= \sum_{a,b} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{i_{N-1} \cdots i_1 b} x_a x_b^* \\ &= \sum_{i_{k+1}} \left(\sum_{a,b} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{i_{N-1} \cdots i_1 b} x_a x_b^* \right) \\ &= \sum_{i_{k+1}} \hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}} . \end{aligned}$	$\begin{aligned} \hat{\rho}_{x_{i_k} \cdots x_{i_1}} &= \sum_{a,b} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{i_{N-1} \cdots i_1 b} x_a x_b^* \\ &= \sum_{i_{k+1}} \left(\sum_{a,b} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{i_{N-1} \cdots i_1 b} x_a x_b^* \right) \\ &= \sum_{i_{k+1}} \hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}} . \end{aligned}$
pdf-dc99cf88_E00016	008	■0.953 ● W +0	$\hat{\rho}_{x_{i_k} \cdots x_{i_1}} = \sum_{i_{k+1}} \hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}$	$\hat{\rho}_{x_{i_k} \cdots x_{i_1}} = \sum_{i_{k+1}} \hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}$	$\hat{\rho}_{x_{i_k} \cdots x_{i_1}} = \sum_{i_{k+1}} \hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}$
pdf-dc99cf88_E00038	014	■0.974 ● C +3	$\rho_{\text{dog}} \geq \pi(\text{black} \mid \text{dog}) \rho_{\text{black dog}} \geq \pi(\text{small black} \mid \text{dog}) \rho_{\text{small black dog}} .$	$\rho_{\text{dog}} \geq \pi(\text{black} \mid \text{dog}) \rho_{\text{black dog}} \geq \pi(\text{small black} \mid \text{dog}) \rho_{\text{small black dog}} .$	$\rho_{\text{dog}} \geq \pi(\text{black} \mid \text{dog}) \rho_{\text{black dog}} \geq \pi(\text{small black} \mid \text{dog}) \rho_{\text{small black dog}} .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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pdf-dc99cf88_E0003 (3)	005	0.990 N +0	$\operatorname{Pr}_{\psi} \{ \psi_i = \psi \mid x_i = y_i \} = \frac{1}{\sum_{j,b} \bar{\psi}_j x_j^* y_b^*} \sum_{j,b} \bar{\psi}_j x_j^* y_b^* \psi_i = \sum_{j,b} \bar{\psi}_j x_j^* y_b^* \psi_i$	$\Pr_{\psi} = \psi \psi^* = \left(\sum_{i,a} \psi_{ia} x_i y_a \right) \left(\sum_{j,b} \bar{\psi}_{jb} x_j^* y_b^* \right) = \sum_{i,a} \psi_{ia} \bar{\psi}_{jb} x_i x_j^* y_a y_b^*.$	$\Pr_{\psi} = \psi \psi^* = \left(\sum_{i,a} \psi_{ia} x_i y_a \right) \left(\sum_{j,b} \bar{\psi}_{jb} x_j^* y_b^* \right) = \sum_{i,a} \psi_{ia} \bar{\psi}_{jb} x_i x_j^* y_a y_b^*.$
pdf-dc99cf88_E00033	012	0.992 N +1	$\frac{1}{3} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \rho_{\text{dog}} = \pi(\text{black} \mid \text{dog}) \rho_{\text{black dog}} + \pi(\text{white} \mid \text{dog}) \rho_{\text{white dog}}$	$\frac{1}{3} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \rho_{\text{dog}} = \pi(\text{black} \mid \text{dog}) \rho_{\text{black dog}} + \pi(\text{white} \mid \text{dog}) \rho_{\text{white dog}}$	$\frac{1}{3} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \rho_{\text{dog}} = \pi(\text{black} \mid \text{dog}) \rho_{\text{black dog}} + \pi(\text{white} \mid \text{dog}) \rho_{\text{white dog}}$
pdf-dc99cf88_E00030	012	0.992 W +0	$\begin{aligned} \rho_{x_{i_k} \cdots x_{i_1}} &= \sum_{i_{k+1}} \pi(x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}} \\ &= \sum_{i_{k+2}, i_{k+1}} \pi(x_{i_{k+2}} x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{k+2}} x_{i_{k+1}} x_{i_k} \cdots x_{i_1}} \\ &= \vdots \\ &= \sum_{i_{N-1}, \dots, i_{k+1}} \pi(x_{i_{N-1}} \cdots x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{N-1}} \cdots x_{i_k} \cdots x_{i_1}} \end{aligned}$	$\rho_{x_{i_k} \cdots x_{i_1}} = \sum_{i_{k+1}} \pi(x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}$	$\rho_{x_{i_k} \cdots x_{i_1}} = \sum_{i_{k+1}} \pi(x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}$
pdf-dc99cf88_E00028	011	0.993 W +0	$\begin{aligned} \rho_{x_{i_k} \cdots x_{i_1}} &= \frac{\hat{\rho}_{x_{i_k} \cdots x_{i_1}}}{\pi(x_{i_k} \cdots x_{i_1})} = \sum_{i_{k+1}} \frac{\pi(x_{i_{k+1}} x_{i_k} \cdots x_{i_1})}{\pi(x_{i_k} \cdots x_{i_1})} \frac{\hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}}{\pi(x_{i_{k+1}} x_{i_k} \cdots x_{i_1})} \\ &= \sum_{i_{k+1}} \pi(x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}} \end{aligned}$	$\rho_{x_{i_k} \cdots x_{i_1}} = \frac{\hat{\rho}_{x_{i_k} \cdots x_{i_1}}}{\pi(x_{i_k} \cdots x_{i_1})} = \sum_{i_{k+1}} \frac{\pi(x_{i_{k+1}} x_{i_k} \cdots x_{i_1})}{\pi(x_{i_k} \cdots x_{i_1})} \frac{\hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}}{\pi(x_{i_{k+1}} x_{i_k} \cdots x_{i_1})}$	$\rho_{x_{i_k} \cdots x_{i_1}} = \frac{\hat{\rho}_{x_{i_k} \cdots x_{i_1}}}{\pi(x_{i_k} \cdots x_{i_1})} = \sum_{i_{k+1}} \frac{\pi(x_{i_{k+1}} x_{i_k} \cdots x_{i_1})}{\pi(x_{i_k} \cdots x_{i_1})} \frac{\hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}}{\pi(x_{i_{k+1}} x_{i_k} \cdots x_{i_1})}$
pdf-dc99cf88_E00036 (10)	013	0.995 W +1	$\hat{\rho}_{\text{dog}} \geq \hat{\rho}_{\text{black dog}} \geq \hat{\rho}_{\text{small black dog}}.$	$\hat{\rho}_{\text{dog}} \geq \hat{\rho}_{\text{black dog}} \geq \hat{\rho}_{\text{small black dog}}.$	$\hat{\rho}_{\text{dog}} \geq \hat{\rho}_{\text{black dog}} \geq \hat{\rho}_{\text{small black dog}}.$
pdf-dc99cf88_E00046	016	0.996 W +2	$\mathcal{D}(\hat{\rho}_s, \hat{\rho}_{s'}) = \begin{cases} \text{tr } \hat{\rho}_{s'} / \text{tr } \hat{\rho}_s & \text{if } s \leq s' \\ 0 & \text{otherwise} \end{cases}$	$\mathcal{D}(\hat{\rho}_s, \hat{\rho}_{s'}) = \begin{cases} \text{tr } \hat{\rho}_{s'} / \text{tr } \hat{\rho}_s & \text{if } s \leq s' \\ 0 & \text{otherwise} \end{cases}$	$\mathcal{D}(\hat{\rho}_s, \hat{\rho}_{s'}) = \begin{cases} \text{tr } \hat{\rho}_{s'} / \text{tr } \hat{\rho}_s & \text{if } s \leq s' \\ 0 & \text{otherwise} \end{cases}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
pdf-dc99cf88_EQ0019	009	■ 0.997 ● W +0	$\psi=\frac{1}{\sqrt{5}}\left(\begin{array}{cccccc}\text{small} & \otimes & \text{black} & \otimes & \text{dog} & \otimes & \text{barks} & + \\ \text{small} & \otimes & \text{white} & \otimes & \text{dog} & \otimes & \text{barks} & + \\ \text{big} & \otimes & \text{black} & \otimes & \text{dog} & \otimes & \text{runs} & + \\ \text{big} & \otimes & \text{white} & \otimes & \text{cat} & \otimes & \text{runs} & + \\ \text{small} & \otimes & \text{black} & \otimes & \text{cat} & \otimes & \text{runs} & \end{array}\right)$	$\psi=\frac{1}{\sqrt{5}}\left(\begin{array}{cccccc}\text{small} & \otimes & \text{black} & \otimes & \text{dog} & \otimes & \text{barks} & + \\ \text{small} & \otimes & \text{white} & \otimes & \text{dog} & \otimes & \text{barks} & + \\ \text{big} & \otimes & \text{black} & \otimes & \text{dog} & \otimes & \text{runs} & + \\ \text{big} & \otimes & \text{white} & \otimes & \text{cat} & \otimes & \text{runs} & + \\ \text{small} & \otimes & \text{black} & \otimes & \text{cat} & \otimes & \text{runs} & \end{array}\right)$	
pdf-dc99cf88_EQ0027	011	■ 0.997 ● N +0	$\rho_{x_{i_k}\cdots x_{i_1}}=\sum_{i_{k+1}}\pi\left(x_{i_{k+1}}\mid x_{i_k}\cdots x_{i_1}\right)\rho_{x_{i_{k+1}}x_{i_k}\cdots x_{i_1}}.$	$\rho_{x_{i_k}\cdots x_{i_1}}=\sum_{i_{k+1}}\pi\left(x_{i_{k+1}}\mid x_{i_k}\cdots x_{i_1}\right)\rho_{x_{i_{k+1}}x_{i_k}\cdots x_{i_1}}.$	$\rho_{x_{i_k}\cdots x_{i_1}}=\sum_{i_{k+1}}\pi(x_{i_{k+1}} x_{i_k}\cdots x_{i_1})\rho_{x_{i_{k+1}}x_{i_k}\cdots x_{i_1}}.$
pdf-dc99cf88_EQ0022	010	■ 0.998 ● N +1	$\hat{\rho}_{\text{black dog}}=\frac{1}{5}\left[\begin{array}{cc}1&0\\0&1\end{array}\right]\quad\hat{\rho}_{\text{small black dog}}=\frac{1}{5}\left[\begin{array}{cc}1&0\\0&0\end{array}\right]$	$\hat{\rho}_{\text{black dog}}=\frac{1}{5}\left[\begin{array}{cc}1&0\\0&1\end{array}\right]\quad\hat{\rho}_{\text{small black dog}}=\frac{1}{5}\left[\begin{array}{cc}1&0\\0&0\end{array}\right]$	$\hat{\rho}_{\text{black dog}}=\frac{1}{5}\left[\begin{array}{cc}1&0\\0&1\end{array}\right]\quad\hat{\rho}_{\text{small black dog}}=\frac{1}{5}\left[\begin{array}{cc}1&0\\0&0\end{array}\right]$
pdf-dc99cf88_EQ0035 (9)	013	■ 0.998 ● N +0	$\hat{\rho}_{x_{i_k}\cdots x_{i_1}}\geq\hat{\rho}_{x_{i_{k+1}}x_{i_k}\cdots x_{i_1}}$	$\hat{\rho}_{x_{i_k}\cdots x_{i_1}}\geq\hat{\rho}_{x_{i_{k+1}}x_{i_k}\cdots x_{i_1}}$	$\hat{\rho}_{x_{i_k}\cdots x_{i_1}}\geq\hat{\rho}_{x_{i_{k+1}}x_{i_k}\cdots x_{i_1}}$
pdf-dc99cf88_EQ0023	011	■ 0.999 ● W +0	$\sum_{i_{N-1},\ldots,i_{k+1},a}\left \left\langle\psi,x_{i_{N-1}}\cdots x_{i_3}x_{i_2}x_{i_1}x_a\right\rangle\right ^2=\sum_{i_{N-1},\ldots,i_{k+1},a}\pi\left(x_{i_{N-1}},\ldots,x_{i_3},x_{i_2},x_{i_1},x_a\right)$	$\sum_{i_{N-1},\ldots,i_{k+1},a}\left \left\langle\psi,x_{i_{N-1}}\cdots x_{i_3}x_{i_2}x_{i_1}x_a\right\rangle\right ^2=\sum_{i_{N-1},\ldots,i_{k+1},a}\pi\left(x_{i_{N-1}},\ldots,x_{i_3},x_{i_2},x_{i_1},x_a\right)$	$\sum_{i_{N-1},\ldots,i_{k+1},a}\left \left\langle\psi,x_{i_{N-1}}\cdots x_{i_3}x_{i_2}x_{i_1}x_a\right\rangle\right ^2=\sum_{i_{N-1},\ldots,i_{k+1},a}\pi\left(x_{i_{N-1}},\ldots,x_{i_3},x_{i_2},x_{i_1},x_a\right)$
pdf-dc99cf88_EQ0010	006	■ 0.999 ● K +0	$\rho_X=\frac{1}{4}\left[\begin{array}{cccc}1&1&0&0\\1&1&0&0\\0&0&1&1\\0&0&1&1\end{array}\right]\quad\rho_Y=\frac{1}{4}\left[\begin{array}{cc}2&0\\0&2\end{array}\right]$	$\rho_X=\frac{1}{4}\left[\begin{array}{cccc}1&1&0&0\\1&1&0&0\\0&0&1&1\\0&0&1&1\end{array}\right]\quad\rho_Y=\frac{1}{4}\left[\begin{array}{cc}2&0\\0&2\end{array}\right]$	$\rho_X=\frac{1}{4}\left[\begin{array}{cccc}1&1&0&0\\1&1&0&0\\0&0&1&1\\0&0&1&1\end{array}\right]\quad\rho_Y=\frac{1}{4}\left[\begin{array}{cc}2&0\\0&2\end{array}\right]$
pdf-dc99cf88_EQ0014	007	■ 0.999 ● N +0	$\hat{\rho}_{x_{i_2}x_{i_1}}:=\sum_{i_4,i_3,a,b}\psi_{i_4i_3i_2i_1a}\bar{\psi}_{i_4i_3i_2i_1b}x_a^*x_b$	$\hat{\rho}_{x_{i_2}x_{i_1}}:=\sum_{i_4,i_3,a,b}\psi_{i_4i_3i_2i_1a}\bar{\psi}_{i_4i_3i_2i_1b}x_a^*x_b$	$\hat{\rho}_{x_{i_2}x_{i_1}}:=\sum_{i_4,i_3,a,b}\psi_{i_4i_3i_2i_1a}\bar{\psi}_{i_4i_3i_2i_1b}x_a^*x_b$
pdf-dc99cf88_EQ0005	005	■ 1.000 ● K +0	$\rho_Y=\left[\begin{array}{cccc}\pi_Y(y_1)&&&*\\&\pi_Y(y_2)&&\\&&\ddots&*&&&\pi_Y(y_m)\end{array}\right]$	$\rho_Y=\left[\begin{array}{cccc}\pi_Y(y_1)&&&*\\&\pi_Y(y_2)&&\\&&\ddots&*&&&\pi_Y(y_m)\end{array}\right]$	$\rho_Y=\left[\begin{array}{cccc}\pi_Y(y_1)&&&*\\&\pi_Y(y_2)&&\\&&\ddots&*&&&\pi_Y(y_m)\end{array}\right]$

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
pdf-dc99cf88_EQ0018 (7)	009	■ 1.000 ● W +0	$\hat{\rho}_{x_{i_1}} = \sum_{i_2} \hat{\rho}_{x_{i_2} x_{i_1}} = \sum_{i_3, i_2} \hat{\rho}_{x_{i_3} x_{i_2} x_{i_1}} = \cdots = \sum_{i_{N-1}, \dots, i_3, i_2} \hat{\rho}_{x_{i_{N-1}} \cdots x_{i_3} x_{i_2} x_{i_1}}$		
pdf-dc99cf88_EQ0026	011	■ 1.000 ● N +0	$\hat{\rho}_{\mathrm{dog}} = \frac{1}{5} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \quad \rho_{\mathrm{dog}} = \frac{1}{3} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$		
pdf-dc99cf88_EQ0040	014	■ 1.000 ● N +0	$(x_{i_{N-1}}, \ldots, \overbrace{x_{i_k}, \ldots, x_{i_1}}^s, x_a).$		
pdf-dc99cf88_EQ0001 (1)	004	■ 1.000 ● N +0	$\psi = \sum_{s \in S} \sqrt{\pi(s)} s$		
pdf-dc99cf88_EQ0002 (2)	005	■ 1.000 ● N +0	$\psi = \sum_{i,a} \psi_{ia} x_i y_a$		
pdf-dc99cf88_EQ0004 (4)	005	■ 1.000 ● W +2	$\rho_Y = \mathrm{tr}_X \mathrm{Pr}_\psi = \sum_{i,a} \psi_{ia} \bar{\psi}_{jb} \mathrm{tr}_X (x_i x_j^* y_a y_b^*) = \sum_{i,a} \psi_{ia} \bar{\psi}_{jb} \mathrm{tr} (x_i x_j^*) \cdot y_a y_b^* = \sum_{a,b} \psi_{ia} \bar{\psi}_{ib} y_a y_b^*$		
pdf-dc99cf88_EQ0006	005	■ 1.000 ● W +0	$\sum_i \sqrt{\pi(x_i y_a) \pi(x_i y_b)} = \frac{1}{\sqrt{\pi_Y(y_a) \pi_Y(y_b)}} \sum_i \sqrt{\pi(x_i, y_a) \pi(x_i, y_b)} = \frac{(\rho_Y)_{ab}}{\sqrt{\pi_Y(y_a) \pi_Y(y_b)}}$		
pdf-dc99cf88_EQ0007	006	■ 1.000 ● W +1	$\rho_Y(y_c) = \sum_{i,a} \sqrt{\pi(x_i, y_a) \pi(x_i, y_c)} y_a = \sum_{i,a} \sqrt{\pi(x_i, y_a) \pi(x_i, y_d)} y_a = \rho_Y(y_d).$		
pdf-dc99cf88_EQ0011 (5)	007	■ 1.000 ● N +0	$\psi = \sum_{i_{N-1}, \dots, i_1, a} \psi_{i_{N-1} \cdots i_1 a} x_{i_{N-1}} \cdots x_{i_1} x_a$		
pdf-dc99cf88_EQ0012	007	■ 1.000 ● N +1	$\mathrm{Pr}_\psi = \psi \psi^* = \sum_{i_{N-1}, \dots, i_1, a} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{j_{N-1} \cdots j_1 b} x_{i_{N-1}} \cdots x_{i_1} x_a x_{j_{N-1}}^* \cdots x_{j_1}^* x_b^*$		
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
pdf-dc99cf88_EQ0013 (6)	007	■ 1.000 ● N +0	$\rho_V=\sum_{i=1}^N a_i b_i \psi_{i_1} \dots \psi_{i_N}$	$\rho_V = \sum_{i_1, \dots, i_N} \psi_{i_1} \dots \psi_{i_N} x_{i_1} \dots x_{i_N}$	$\rho_V = \sum_{i_1, \dots, i_N} \psi_{i_1} \dots \psi_{i_N} x_{i_1} \dots x_{i_N}$
pdf-dc99cf88_EQ0024	011	■ 1.000 ● N +0	$\rho_{x_{i_1} \dots x_{i_N}} := \hat{\rho}_{x_{i_1} \dots x_{i_N}} / \pi(x_{i_1} \dots x_{i_N})$	$\rho_{x_{i_1} \dots x_{i_N}} := \hat{\rho}_{x_{i_1} \dots x_{i_N}} / \pi(x_{i_1} \dots x_{i_N})$	$\rho_{x_{i_1} \dots x_{i_N}} := \hat{\rho}_{x_{i_1} \dots x_{i_N}} / \pi(x_{i_1} \dots x_{i_N})$
pdf-dc99cf88_EQ0025	011	■ 1.000 ● W +0	$\frac{1}{\pi(x_{i_1} \dots x_{i_N})} \sum_{i_1, \dots, i_N} \langle \psi, x_{i_1} \dots x_{i_N} \rangle ^2 = \frac{\pi(x_{i_1} \dots x_{i_N})}{\pi(x_{i_1} \dots x_{i_N})} = \pi(x_{i_1} \dots x_{i_N})$	$\frac{1}{\pi(x_{i_1} \dots x_{i_N})} \sum_{i_1, \dots, i_N} \langle \psi, x_{i_1} \dots x_{i_N} \rangle ^2 = \frac{\pi(x_{i_1} \dots x_{i_N})}{\pi(x_{i_1} \dots x_{i_N})} = \pi(x_{i_1} \dots x_{i_N})$	$\frac{1}{\pi(x_{i_1} \dots x_{i_N})} \sum_{i_1, \dots, i_N} \langle \psi, x_{i_1} \dots x_{i_N} \rangle ^2 = \frac{\pi(x_{i_1} \dots x_{i_N})}{\pi(x_{i_1} \dots x_{i_N})} = \pi(x_{i_1} \dots x_{i_N})$
pdf-dc99cf88_EQ0029 (8)	012	■ 1.000 ● N +0	$\rho_{x_{i_1} \dots x_{i_N}} = \sum_{i_1, \dots, i_N} \pi(x_{i_1} \dots x_{i_N} x_{i_1}) \rho_{x_{i_1} \dots x_{i_N} x_{i_1}}$	$\rho_{x_{i_1}} = \sum_{i_1, \dots, i_N} \pi(x_{i_1} \dots x_{i_N} x_{i_1}) \rho_{x_{i_1} \dots x_{i_N} x_{i_1}}$	$\rho_{x_{i_1}} = \sum_{i_1, \dots, i_N} \pi(x_{i_1} \dots x_{i_N} x_{i_1}) \rho_{x_{i_1} \dots x_{i_N} x_{i_1}}$
pdf-dc99cf88_EQ0032	012	■ 1.000 ● N +1	$\rho_{\text{black dog}} = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \rho_{\text{white dog}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\rho_{\text{black dog}} = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \rho_{\text{white dog}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\rho_{\text{black dog}} = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \rho_{\text{white dog}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$
pdf-dc99cf88_EQ0037 (11)	014	■ 1.000 ● N +1	$\rho_{x_{i_1} \dots x_{i_N}} \geq \pi(x_{i_1} \dots x_{i_N} x_{i_1}) \rho_{x_{i_1} \dots x_{i_N} x_{i_1}}$	$\rho_{x_{i_1} \dots x_{i_N}} \geq \pi(x_{i_1} \dots x_{i_N} x_{i_1}) \rho_{x_{i_1} \dots x_{i_N} x_{i_1}}$	$\rho_{x_{i_1} \dots x_{i_N}} \geq \pi(x_{i_1} \dots x_{i_N} x_{i_1}) \rho_{x_{i_1} \dots x_{i_N} x_{i_1}}$
pdf-dc99cf88_EQ0041	015	■ 1.000 ● W +0	$\left\{ x_{i_1} \operatorname{dog} \mid x_{i_1} \in X \right\} \sqcup \left\{ x_{i_2} \operatorname{dog} \mid x_{i_2} \in X \right\} \sqcup \dots \sqcup \left\{ x_{i_N} \operatorname{dog} \mid x_{i_N} \in X \right\}$	$\left\{ x_{i_1} \operatorname{dog} \mid x_{i_1} \in X \right\} \sqcup \left\{ x_{i_2} \operatorname{dog} \mid x_{i_2} \in X \right\} \sqcup \dots \sqcup \left\{ x_{i_N} \operatorname{dog} \mid x_{i_N} \in X \right\}$	$\left\{ x_{i_1} \operatorname{dog} \mid x_{i_1} \in X \right\} \sqcup \left\{ x_{i_2} \operatorname{dog} \mid x_{i_2} \in X \right\} \sqcup \dots \sqcup \left\{ x_{i_N} \operatorname{dog} \mid x_{i_N} \in X \right\}$
pdf-dc99cf88_EQ0044	016	■ 1.000 ● N +0	$\mathcal{C}(c, c') \leq \mathcal{D}(fc, fc')$	$\mathcal{C}(c, c') \leq \mathcal{D}(fc, fc')$	$\mathcal{C}(c, c') \leq \mathcal{D}(fc, fc')$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered
pdf-dc99cf88_F00001	001	—	{ }^{+}	+
pdf-dc99cf88_F00002	003	—	s	s
pdf-dc99cf88_F00003	003	—	\rho_{s}	ρ_s
pdf-dc99cf88_F00004	003	—	s \mapsto \rho_{s}	$s \mapsto \rho_s$
pdf-dc99cf88_F00005	003	—	\leq	$\leq$
pdf-dc99cf88_F00006	003	—	\hat{\rho}_{s}	$\hat{\rho}_s$
pdf-dc99cf88_F00007	003	—	s \mapsto \hat{\rho}_{s}	$s \mapsto \hat{\rho}_s$
pdf-dc99cf88_F00008	004	—	S	S
pdf-dc99cf88_F00009	004	—	V=\mathbb{C}^{\{S\}}	$V = \mathbb{C}^S$
pdf-dc99cf88_F00010	004	—	V	V
pdf-dc99cf88_F00011	004	—	s \in S	$s \in S$
pdf-dc99cf88_F00012	004	—	s \in V	$s \in V$
pdf-dc99cf88_F00013	004	—	S=\left\{s_1\right\}, \ldots, s_d\right\}	$S=\left\{s_1, \ldots, s_d\right\}$
pdf-dc99cf88_F00014	004	—	s_{i}	s_i
pdf-dc99cf88_F00015	004	—	i	i
pdf-dc99cf88_F00016	004	—	\mathbb{C}^d	$\mathbb{C}^d$
pdf-dc99cf88_F00017	004	—	V \cong \mathbb{C}^d	$V \cong \mathbb{C}^d$
pdf-dc99cf88_F00018	004	—	\left\langle s_i, s_j \right\rangle	$\left\langle s_i, s_j \right\rangle$
pdf-dc99cf88_F00019	004	—	i=j	$i=j$
pdf-dc99cf88_F00020	004	—	v \in V	$v \in V$
pdf-dc99cf88_F00021	004	—	v^{\ast}: V \rightarrow \mathbb{C}	$v^{\ast}: V \rightarrow \mathbb{C}$
pdf-dc99cf88_F00022	004	—	v^{\prime} \mapsto \left\langle v, v^{\prime} \right\rangle	$v^{\prime} \mapsto \left\langle v, v^{\prime} \right\rangle$
pdf-dc99cf88_F00023	004	—	V^{\ast}:=\operatorname{hom}(V, \mathbb{C})	$V^{\ast}:=\operatorname{hom}(V, \mathbb{C})$
pdf-dc99cf88_F00024	004	—	W	W
pdf-dc99cf88_F00025	004	—	V \otimes W	$V \otimes W$
pdf-dc99cf88_F00026	004	—	v w	vw
pdf-dc99cf88_F00027	004	—	v \otimes w	$v \otimes w$
pdf-dc99cf88_F00028	004	—	w v^{\ast}	$w v^{\ast}$
pdf-dc99cf88_F00029	004	—	W \otimes V^{\ast}	$W \otimes V^{\ast}$
pdf-dc99cf88_F00030	004	—	V \rightarrow W	$V \rightarrow W$
pdf-dc99cf88_F00031	004	—	v^{\prime} \mapsto w \left\langle v, v^{\prime} \right\rangle	$v^{\prime} \mapsto w \left\langle v, v^{\prime} \right\rangle$
pdf-dc99cf88_F00032	004	—	\operatorname{End}(V)	$\operatorname{End}(V)$
pdf-dc99cf88_F00033	004	—	\psi \in V	$\psi \in V$
pdf-dc99cf88_F00034	004	—	\psi \psi^{\ast} \in V \otimes V^{\ast}	$\psi \psi^{\ast} \in V \otimes V^{\ast}$
pdf-dc99cf88_F00035	004	—	\psi	ψ
pdf-dc99cf88_F00036	004	—	\operatorname{Pr}_{\psi}	Pr_{ψ}
pdf-dc99cf88_F00037	004	—	\rho	ρ
pdf-dc99cf88_F00038	004	—	\rho \geq 0	$\rho \geq 0$
pdf-dc99cf88_F00039	004	—	\pi_{\rho}: S \rightarrow \mathbb{R}	$\pi_{\rho}: S \rightarrow \mathbb{R}$
pdf-dc99cf88_F00040	004	—	\pi_{\rho}(s):=\left\langle \rho s, s \right\rangle	$\pi_{\rho}(s):=\left\langle \rho s, s \right\rangle$
pdf-dc99cf88_F00041	004	—	\pi: S \rightarrow \mathbb{R}	$\pi: S \rightarrow \mathbb{R}$
pdf-dc99cf88_F00042	004	—	\pi	π
pdf-dc99cf88_F00043	004	—	\sum_s \pi(s) s s^{\ast}	$\sum_s \pi(s) s s^{\ast}$
pdf-dc99cf88_F00044	004	—	\pi(s)	$\pi(s)$
pdf-dc99cf88_F00045	004	—	\operatorname{Pr}_{\psi}: V \rightarrow V	$\operatorname{Pr}_{\psi}: V \rightarrow V$
pdf-dc99cf88_F00046	004	—	\pi_{\operatorname{Pr}_{\psi}}(s)=\left\langle \operatorname{Pr}_{\psi} s, s \right\rangle=\sqrt{\pi(s)^2}=\pi(s)	$\pi_{\operatorname{Pr}_{\psi}}(s)=\left\langle \operatorname{Pr}_{\psi} s, s \right\rangle=\sqrt{\pi(s)^2}=\pi(s)$
pdf-dc99cf88_F00047	004	—	\pi \mapsto \operatorname{Pr}_{\psi}	$\pi \mapsto \operatorname{Pr}_{\psi}$
pdf-dc99cf88_F00048	004	—	A	A
pdf-dc99cf88_F00049	004	—	B	B
pdf-dc99cf88_F00050	004	—	\operatorname{End}(A \otimes B) \cong \operatorname{End}(A) \otimes	$\operatorname{End}(A \otimes B) \cong \operatorname{End}(A) \otimes$
pdf-dc99cf88_F00051	004	—	\operatorname{End}(B)	$\operatorname{End}(B)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value				
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
pdf-dc99cf88_F00052	004	—	$\operatorname{tr}_{\{A\}} := \operatorname{tr} \otimes \operatorname{id}_{\{B\}}$	$\operatorname{tr}_A := \operatorname{tr} \otimes \operatorname{id}_B$
pdf-dc99cf88_F00053	004	—	$\operatorname{tr}_{\{B\}} := \operatorname{tr}_{\operatorname{id}_{\{A\}}} \otimes \operatorname{tr}$	$\operatorname{tr}_B := \operatorname{id}_A \otimes \operatorname{tr}$
pdf-dc99cf88_F00054	004	—	$\rho: A \otimes B \rightarrow$	$\rho: A \otimes B \rightarrow$
pdf-dc99cf88_F00055	004	—	$A \otimes B$	$A \otimes B$
pdf-dc99cf88_F00056	004	—	$\rho_{\{B\}} := \operatorname{tr}_{\{A\}} \rho: B \rightarrow B$	$\rho_B := \operatorname{tr}_A \rho: B \rightarrow B$
pdf-dc99cf88_F00057	004	—	$\rho_{\{A\}} := \operatorname{tr}_{\{B\}} \rho: A \rightarrow A$	$\rho_A := \operatorname{tr}_B \rho: A \rightarrow A$
pdf-dc99cf88_F00058	004	—	$S = X \times Y$	$S = X \times Y$
pdf-dc99cf88_F00059	004	—	$X = \{x_1, \ldots, x_n\}$	$X = \{x_1, \ldots, x_n\}$
pdf-dc99cf88_F00060	004	—	$Y = \{y_1, \ldots, y_m\}$	$Y = \{y_1, \ldots, y_m\}$
pdf-dc99cf88_F00061	004	—	$\operatorname{Pr}_{\psi}: \mathbb{C}^X \otimes \mathbb{C}^Y \rightarrow \mathbb{C}^X \otimes \mathbb{C}^Y$	$\operatorname{Pr}_{\psi}: \mathbb{C}^X \otimes \mathbb{C}^Y \rightarrow \mathbb{C}^X \otimes \mathbb{C}^Y$
pdf-dc99cf88_F00062	005	—	$\psi_{ia} = \sqrt{\pi} \left(x_i, y_a \right)$	$\psi_{ia} = \sqrt{\pi} \left(x_i, y_a \right)$
pdf-dc99cf88_F00063	005	—	$\rho_Y: \mathbb{C}^Y \rightarrow \mathbb{C}^Y$	$\rho_Y: \mathbb{C}^Y \rightarrow \mathbb{C}^Y$
pdf-dc99cf88_F00064	005	—	$\operatorname{tr} \left(x_i x_j^* \right) = \left\langle x_j, x_i \right\rangle$	$\operatorname{tr} \left(x_i x_j^* \right) = \left\langle x_j, x_i \right\rangle$
pdf-dc99cf88_F00065	005	—	a	a
pdf-dc99cf88_F00066	005	—	ρ_Y	ρ_Y
pdf-dc99cf88_F00067	005	—	$\pi_Y \left(y_a \right) := \sum_i \psi_{ia} \overline{\psi_{ia}} = \sum_i \pi \left(x_i, y_a \right)$	$\pi_Y \left(y_a \right) := \sum_i \psi_{ia} \overline{\psi_{ia}} = \sum_i \pi \left(x_i, y_a \right)$
pdf-dc99cf88_F00068	005	—	$\pi_Y: Y \rightarrow \mathbb{R}$	$\pi_Y: Y \rightarrow \mathbb{R}$
pdf-dc99cf88_F00069	005	—	ab	ab
pdf-dc99cf88_F00070	005	—	$\left(\rho_Y \right)_{ab} = \sum_i \sqrt{\pi \left(x_i, y_a \right) \pi \left(x_i, y_b \right)}$	$\left(\rho_Y \right)_{ab} = \sum_i \sqrt{\pi \left(x_i, y_a \right) \pi \left(x_i, y_b \right)}$
pdf-dc99cf88_F00071	005	—	y_a	y_a
pdf-dc99cf88_F00072	005	—	y_b	y_b
pdf-dc99cf88_F00073	005	—	X	X
pdf-dc99cf88_F00074	005	—	$p, q: S \rightarrow \mathbb{R}$	$p, q: S \rightarrow \mathbb{R}$
pdf-dc99cf88_F00075	005	—	$B(p, q) := \sum_s \sqrt{p(s) q(s)}$	$B(p, q) := \sum_s \sqrt{p(s) q(s)}$
pdf-dc99cf88_F00076	005	—	$y_a \in Y$	$y_a \in Y$
pdf-dc99cf88_F00077	005	—	$\pi_a: X \rightarrow \mathbb{R}$	$\pi_a: X \rightarrow \mathbb{R}$
pdf-dc99cf88_F00078	005	—	$x_i \mapsto \pi \left(x_i \mid y_a \right)$	$x_i \mapsto \pi \left(x_i \mid y_a \right)$
pdf-dc99cf88_F00079	005	—	π_a	π_a
pdf-dc99cf88_F00080	005	—	π_b	π_b
pdf-dc99cf88_F00081	005	—	$\left(\rho_Y \right)_{ab} = \sqrt{\pi_Y \left(y_a \right) \pi_Y \left(y_b \right)} B \left(\pi_a, \pi_b \right)$	$\left(\rho_Y \right)_{ab} = \sqrt{\pi_Y \left(y_a \right) \pi_Y \left(y_b \right)} B \left(\pi_a, \pi_b \right)$
pdf-dc99cf88_F00082	005	—	$H(p, q) := \sqrt{1 - B(p, q)}$	$H(p, q) := \sqrt{1 - B(p, q)}$
pdf-dc99cf88_F00083	005	—	$\rho_X = \operatorname{tr}_Y \operatorname{Pr}_{\psi}$	$\rho_X = \operatorname{tr}_Y \operatorname{Pr}_{\psi}$
pdf-dc99cf88_F00084	005	—	$\pi_X: X \rightarrow \mathbb{R}$	$\pi_X: X \rightarrow \mathbb{R}$
pdf-dc99cf88_F00085	005	—	$(x, y) \in X \times Y$	$(x, y) \in X \times Y$
pdf-dc99cf88_F00086	005	—	x	x
pdf-dc99cf88_F00087	005	—	y	y
pdf-dc99cf88_F00088	005	—	ρ_X	ρ_X
pdf-dc99cf88_F00089	006	—	$\pi: X \times Y \rightarrow \mathbb{R}$	$\pi: X \times Y \rightarrow \mathbb{R}$
pdf-dc99cf88_F00090	006	—	y_c	y_c
pdf-dc99cf88_F00091	006	—	y_d	y_d
pdf-dc99cf88_F00092	006	—	$\pi \left(x_i, y_c \right) = \pi \left(x_i, y_d \right)$	$\pi \left(x_i, y_c \right) = \pi \left(x_i, y_d \right)$
pdf-dc99cf88_F00093	006	—	$\rho_Y = \operatorname{tr}_X Pr_{\psi}$	$\rho_Y = \operatorname{tr}_X Pr_{\psi}$
pdf-dc99cf88_F00094	006	—	$\pi \left(x_i, y_c \right) \neq \pi \left(x_i, y_d \right)$	$\pi \left(x_i, y_c \right) \neq \pi \left(x_i, y_d \right)$
pdf-dc99cf88_F00095	006	—	$\rho_Y \left(y_c \right) \neq \rho_Y \left(y_d \right)$	$\rho_Y \left(y_c \right) \neq \rho_Y \left(y_d \right)$
pdf-dc99cf88_F00096		—	$\square$	$\square$
pdf-dc99cf88_F00097	006	—	$\rho_Y \left(y_a \right)$	$\rho_Y \left(y_a \right)$
pdf-dc99cf88_F00098	006	—	$X = \{$	$X = \{$
pdf-dc99cf88_F00099	006	—	$\}$	$\}$
pdf-dc99cf88_F00100	006	—	$Y = \{$	$Y = \{$
pdf-dc99cf88_F00101	006	—	$T \subseteq X \times Y$	$T \subseteq X \times Y$
pdf-dc99cf88_F00102	006	—	T	T
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
pdf-dc99cf88_F00103	006	—	$\pi(x, y)$	$\pi(x, y)$
pdf-dc99cf88_F00104	006	—	$1 / 4$	$1/4$
pdf-dc99cf88_F00105	006	—	$(x, y) \in T$	$(x, y) \in T$
pdf-dc99cf88_F00106	006	—	$\neq \rho_Y$	$\neq \rho_Y$
pdf-dc99cf88_F00107	006	—	$=\rho_X$	$= \rho_X$
pdf-dc99cf88_F00108	006	—	$\pi(x, y)=\frac{1}{ T }$	$\pi(x, y) = \frac{1}{ T }$
pdf-dc99cf88_F00109	006	—	$\pi(x, y)=0$	$\pi(x, y) = 0$
pdf-dc99cf88_F00110	006	—	$\psi=\frac{1}{\sqrt{ T }} \sum_{(x, y) \in T} x y$	$\psi = \frac{1}{\sqrt{ T }} \sum_{(x, y) \in T} x y$
pdf-dc99cf88_F00111	006	—	$\rho_Y=\operatorname{tr}_X \operatorname{Pr}_{\psi}$	$\rho_Y = \operatorname{tr}_X \operatorname{Pr}_{\psi}$
pdf-dc99cf88_F00112	006	—	$\sum_i \psi_{ia} \psi_{ib} = d/ T $	$\sum_i \psi_{ia} \psi_{ib} = d/ T $
pdf-dc99cf88_F00113	006	—	d	d
pdf-dc99cf88_F00114	006	—	$x_i \in X$	$x_i \in X$
pdf-dc99cf88_F00115	006	—	$(x_i, y_a) \in T$	$(x_i, y_a) \in T$
pdf-dc99cf88_F00116	006	—	$(x_i, y_b) \in T$	$(x_i, y_b) \in T$
pdf-dc99cf88_F00117		—	Y	Y
pdf-dc99cf88_F00118		—	$\mathbb{C}^Y \cong \mathbb{C}^2$	$\mathbb{C}^Y \cong \mathbb{C}^2$
pdf-dc99cf88_F00119		—	$=\left[\begin{smallmatrix} 1 & 0 \end{smallmatrix}\right]^{\top}$	$= \begin{bmatrix} 1 & 0 \end{bmatrix}^{\top}$
pdf-dc99cf88_F00120		—	$=\left[\begin{smallmatrix} 0 & 1 \end{smallmatrix}\right]^{\top}$	$= \begin{bmatrix} 0 & 1 \end{bmatrix}^{\top}$
pdf-dc99cf88_F00121		—	$=\left[\begin{smallmatrix} 1 & 0 & 0 & 0 \end{smallmatrix}\right]^{\top}$	$= \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}^{\top}$
pdf-dc99cf88_F00122	007	—	N	N
pdf-dc99cf88_F00123	007	—	$N \geq 2$	$N \geq 2$
pdf-dc99cf88_F00124	007	—	$S=X^{N-1} \times X$	$S = X^{N-1} \times X$
pdf-dc99cf88_F00125	007	—	$\left(x_{i_{N-1}}, \ldots, x_{i_2}\right), x_{i_1}\right)$	$\left(x_{i_{N-1}}, \ldots, x_{i_2}, x_{i_1}\right)$
pdf-dc99cf88_F00126	007	—	x_a	x_a
pdf-dc99cf88_F00127	007	—	$s=\left(x_{i_{N-1}} \cdots x_{i_2} x_{i_1}\right), x_a\right)$	$s = \left(x_{i_{N-1}} \cdots x_{i_2} x_{i_1}, x_a\right)$
pdf-dc99cf88_F00128	007	—	$x_{i_{N-1}} \cdots x_{i_2} x_{i_1}$	$x_{i_{N-1}} \cdots x_{i_2} x_{i_1}$
pdf-dc99cf88_F00129	007	—	i_1	i_1
pdf-dc99cf88_F00130	007	—	i_{N-1}	i_{N-1}
pdf-dc99cf88_F00131	007	—	$i, j, \ldots$	$i, j, \ldots$
pdf-dc99cf88_F00132	007	—	$a, b, \ldots$	$a, b, \ldots$
pdf-dc99cf88_F00133	007	—	X^k	X^k
pdf-dc99cf88_F00134	007	—	$k \geq 1$	$k \geq 1$
pdf-dc99cf88_F00135	007	—	$k=1$	$k = 1$
pdf-dc99cf88_F00136	007	—	$\psi=\sum_{s \in S} \psi_s s$	$\psi = \sum_{s \in S} \psi_s s$
pdf-dc99cf88_F00137	007	—	$\psi_s=\sqrt{\pi(s)}$	$\psi_s = \sqrt{\pi(s)}$
pdf-dc99cf88_F00138	007	—	$\mathbb{C}^S \cong V^{\otimes N-1} \otimes V$	$\mathbb{C}^S \cong V^{\otimes N-1} \otimes V$
pdf-dc99cf88_F00139	007	—	$V=\mathbb{C}^X$	$V = \mathbb{C}^X$
pdf-dc99cf88_F00140	007	—	$s=x_{i_{N-1}} \cdots x_{i_1} x_a$	$s = x_{i_{N-1}} \cdots x_{i_1} x_a$
pdf-dc99cf88_F00141	007	—	$\psi_{i_{N-1} \cdots i_1 a}=\sqrt{\pi\left(x_{i_{N-1}}, \ldots, x_{i_1}\right), x_a\right)}$	$\psi_{i_{N-1} \cdots i_1 a} = \sqrt{\pi\left(x_{i_{N-1}}, \ldots, x_{i_1}, x_a\right)}$
pdf-dc99cf88_F00142	007	—	$V^{\otimes N-1} \otimes V$	$V^{\otimes N-1} \otimes V$
pdf-dc99cf88_F00143	007	—	$\rho_V=\operatorname{tr}_{V^{\otimes N-1}} \operatorname{Pr}_{\psi}: V \rightarrow V$	$\rho_V = \operatorname{tr}_{V^{\otimes N-1}} \operatorname{Pr}_{\psi}: V \rightarrow V$
pdf-dc99cf88_F00144	007	—	X^{N-1}	X^{N-1}
pdf-dc99cf88_F00145	007	—	$N=5$	$N = 5$
pdf-dc99cf88_F00146	007	—	$x_{i_2} x_{i_1}$	$x_{i_2} x_{i_1}$
pdf-dc99cf88_F00147	007	—	i_2	i_2
pdf-dc99cf88_F00148	008	—	i_3	i_3
pdf-dc99cf88_F00149	008	—	i_4	i_4
pdf-dc99cf88_F00150	008	—	ρ_V	ρ_V
pdf-dc99cf88_F00151	008	—	$\hat{\rho}_{x_i x_{i_1}}$	$\hat{\rho}_{x_i x_{i_1}}$
pdf-dc99cf88_F00152	008	—	$\hat{\rho}_{x_{i_2} x_{i_1}}$	$\hat{\rho}_{x_{i_2} x_{i_1}}$
pdf-dc99cf88_F00153	008	—	$\rho_{x_{i_2} x_{i_1}}$	$\rho_{x_{i_2} x_{i_1}}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
pdf-dc99cf88_F00154	008	—	$\hat{\rho}_{x_{i_k} \cdots x_{i_1}}$	$\hat{\rho}_{x_{i_k} \cdots x_{i_1}}$
pdf-dc99cf88_F00155	008	—	$x_{i_k} \cdots x_{i_1}$	$x_{i_k} \cdots x_{i_1}$
pdf-dc99cf88_F00156	008	—	x_{i_1}	x_{i_1}
pdf-dc99cf88_F00157	008	—	$x_{i_1}^{\prime}$	x_{i_1}'
pdf-dc99cf88_F00158	008	—	$\hat{\rho}_{x_{i_1}} = \hat{\rho}_{x_{i_1}^{\prime}}$	$\hat{\rho}_{x_{i_1}} = \hat{\rho}_{x_{i_1}'}$
pdf-dc99cf88_F00159	008	—	$\psi_{i_{N-1} \cdots i_1} a = \psi_{i_{N-1} \cdots i_1}^{\prime} a$	$\psi_{i_{N-1} \cdots i_1} a = \psi_{i_{N-1} \cdots i_1'} a$
pdf-dc99cf88_F00160	008	—	$i_{N-1}, \ldots, i_2$	$i_{N-1}, \dots, i_2$
pdf-dc99cf88_F00161	008	—	$1 \leq k \leq N-1$	$1 \leq k \leq N-1$
pdf-dc99cf88_F00162	008	—	$x_{i_1} = \text{dog}$	$x_{i_1} = \text{dog}$
pdf-dc99cf88_F00163	009	—	$x_{i_{N-1}} \cdots x_{i_1}$	$x_{i_{N-1}} \cdots x_{i_1}$
pdf-dc99cf88_F00164	009	—	$\hat{\rho}_{x_{i_{N-1}} \cdots x_{i_3} x_{i_2} x_{i_1}}$	$\hat{\rho}_{x_{i_{N-1}} \cdots x_{i_3} x_{i_2} x_{i_1}}$
pdf-dc99cf88_F00165	009	—	$A \times B \times C \times D$	$A \times B \times C \times D$
pdf-dc99cf88_F00166	009	—	$X \times X \times X \times X$	$X \times X \times X \times X$
pdf-dc99cf88_F00167	009	—	2×2	2×2
pdf-dc99cf88_F00168	009	—	$A = \{$	$A = \{$
pdf-dc99cf88_F00169	009	—	$B = \{$	$B = \{$
pdf-dc99cf88_F00170	009	—	$C = \{\mathrm{dog}$	$C = \{\text{dog}$
pdf-dc99cf88_F00171	009	—	$D =$	$D =$
pdf-dc99cf88_F00172	009	—	$N = 4$	$N = 4$
pdf-dc99cf88_F00173	009	—	$\pi: A \times B \times C \times D \rightarrow \mathbb{R}$	$\pi: A \times B \times C \times D \rightarrow \mathbb{R}$
pdf-dc99cf88_F00174	009	—	$T \subseteq A \times B \times C \times D$	$T \subseteq A \times B \times C \times D$
pdf-dc99cf88_F00175	009	—	$\mathbb{C}^A \otimes \mathbb{C}^B \otimes \mathbb{C}^C \otimes \mathbb{C}^D$	$\mathbb{C}^A \otimes \mathbb{C}^B \otimes \mathbb{C}^C \otimes \mathbb{C}^D$
pdf-dc99cf88_F00176	009	—	$\mathbb{C}^2$	$\mathbb{C}^2$
pdf-dc99cf88_F00177	009	—	$A \times B \times C$	$A \times B \times C$
pdf-dc99cf88_F00178	009	—	D	D
pdf-dc99cf88_F00179	009	—	$\rho_D = \operatorname{tr}_{A \times B \times C} \operatorname{Pr}_{\psi}$	$\rho_D = \operatorname{tr}_{A \times B \times C} \operatorname{Pr}_{\psi}$
pdf-dc99cf88_F00180	017	—	$\hat{\rho}_{x_{i_1}}$	$\hat{\rho}_{x_{i_1}}$
pdf-dc99cf88_F00181		—	$V^{\otimes N-1} \rightarrow V$	$V^{\otimes N-1} \rightarrow V$
pdf-dc99cf88_F00182	014	—	$\psi \in V^{\otimes N-1} \otimes V$	$\psi \in V^{\otimes N-1} \otimes V$
pdf-dc99cf88_F00183	010	—	$\left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right]$	$\left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right]$
pdf-dc99cf88_F00184	010	—	$\left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right]$	$\left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right]$
pdf-dc99cf88_F00185	010	—	$1/5$	$1/5$
pdf-dc99cf88_F00186	010	—	ρ_D	ρ_D
pdf-dc99cf88_F00187	010	—	$(a, b, c) \in T$	$(a, b, c) \in T$
pdf-dc99cf88_F00188	010	—	$\hat{\rho}_{\text{dog}}$	$\hat{\rho}_{\text{dog}}$
pdf-dc99cf88_F00189	010	—	dog	dog
pdf-dc99cf88_F00190	010	—	$c = \text{dog}$	$c = \text{dog}$
pdf-dc99cf88_F00191	010	—	$(a, b) \in A \times B$	$(a, b) \in A \times B$
pdf-dc99cf88_F00192	010	—	(a, b)	(a, b)
pdf-dc99cf88_F00193	010	—	$b =$	$b =$
pdf-dc99cf88_F00194	010	—	$a \in A$	$a \in A$
pdf-dc99cf88_F00195	010	—	$a =$	$a =$
pdf-dc99cf88_F00196	010	—	dog	dog
pdf-dc99cf88_F00197	010	—	$\hat{\rho}_{\text{dog}} = \hat{\rho}_{\text{black dog}} + \hat{\rho}_{\text{white dog}}$	$\hat{\rho}_{\text{dog}} = \hat{\rho}_{\text{black dog}} + \hat{\rho}_{\text{white dog}}$
pdf-dc99cf88_F00198	010	—	$\hat{\rho}_{\text{black dog}} =$	$\hat{\rho}_{\text{black dog}} =$
pdf-dc99cf88_F00199	010	—	$\hat{\rho}_{\text{small black dog}} + \hat{\rho}_{\text{big black dog}}$	$\hat{\rho}_{\text{small black dog}} + \hat{\rho}_{\text{big black dog}}$
pdf-dc99cf88_F00200	010	—	$\hat{\rho}_{\text{white dog}} = \hat{\rho}_{\text{small white dog}}$	$\hat{\rho}_{\text{white dog}} = \hat{\rho}_{\text{small white dog}}$
pdf-dc99cf88_F00201	011	—	$\pi(x_{i_k}, \ldots, x_{i_1})$	$\pi(x_{i_k}, \dots, x_{i_1})$
pdf-dc99cf88_F00202	011	—	$T \subseteq X^{N-1} \times X$	$T \subseteq X^{N-1} \times X$
pdf-dc99cf88_F00203	011	—	$ T $	$ T $
pdf-dc99cf88_F00204	011	—	$\pi(x_{i_k} \cdots x_{i_1} x_a \mid x_{i_k} \cdots x_{i_1})$	$\pi(x_{i_k} \cdots x_{i_1} x_a \mid x_{i_k} \cdots x_{i_1})$
pdf-dc99cf88_F00205	011	—	$x_a \in X$	x_a

Identifier	Page	Conf.	LaTeX source	Rendered
pdf-dc99cf88_F00207	011	—	$\left\langle \rho_{x_{i_k} \cdots x_{i_1}} x_a, x_a \right\rangle$	$\left\langle \rho_{x_{i_k} \cdots x_{i_1}} x_a, x_a \right\rangle$
pdf-dc99cf88_F00208	011	—	$\operatorname{tr} \hat{\rho}_{\text{dog}} = 3/5 = \pi(\text{dog}) = \sum_{a,b,d} \pi(a,b,\text{dog},d)$	$\operatorname{tr} \hat{\rho}_{\text{dog}} = 3/5 = \pi(\text{dog}) = \sum_{a,b,d} \pi(a,b,\text{dog},d)$
pdf-dc99cf88_F00209	011	—	ρ_{dog}	ρ_{dog}
pdf-dc99cf88_F00210	011	—	$\left\langle \rho_{\text{dog}} \right\rangle$	$\left\langle \rho_{\text{dog}} \right\rangle$
pdf-dc99cf88_F00211	011	—	$\angle = 2/3 = \pi$	$\rangle = 2/3 = \pi$
pdf-dc99cf88_F00212	011	—	$\mid$	$\mid$
pdf-dc99cf88_F00213	011	—	$(a,b,c,d) \in T$	$(a,b,c,d) \in T$
pdf-dc99cf88_F00214	011	—	$d = \text{barks}$	$d = \text{barks}$
pdf-dc99cf88_F00215	011	—	$\angle = 1/3 = \pi$	$\rangle = 1/3 = \pi$
pdf-dc99cf88_F00216	011	—	$\mid \operatorname{dog}$	$\mid \operatorname{dog}$
pdf-dc99cf88_F00217	011	—	ρ_{dog}	ρ_{dog}
pdf-dc99cf88_F00218	011	—	$\rho_{\text{black dog}}$	$\rho_{\text{black dog}}$
pdf-dc99cf88_F00219	011	—	$\rho_{\text{white dog}}$	$\rho_{\text{white dog}}$
pdf-dc99cf88_F00220	011	—	$k+1$	$k+1$
pdf-dc99cf88_F00221	011	—	$\hat{\rho}_{x_{i_k} \cdots x_{i_1}} = \sum_{i_{k+1}} \hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}$	$\hat{\rho}_{x_{i_k} \cdots x_{i_1}} = \sum_{i_{k+1}} \hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}$
pdf-dc99cf88_F00222	012	—	$N-1$	$N-1$
pdf-dc99cf88_F00223	012	—	$\rho_{x_{i_{N-1}} \cdots x_{i_3} x_{i_2} x_{i_1}}$	$\rho_{x_{i_{N-1}} \cdots x_{i_3} x_{i_2} x_{i_1}}$
pdf-dc99cf88_F00224	012	—	$\rho_{x_{i_1}}$	$\rho_{x_{i_1}}$
pdf-dc99cf88_F00225	012	—	$\pi(\mid \operatorname{dog}) = 1/3$	$\pi(\mid \operatorname{dog}) = 1/3$
pdf-dc99cf88_F00226	012	—	$\mid \operatorname{dog}) = 2/3$	$\mid \operatorname{dog}) = 2/3$
pdf-dc99cf88_F00227	012	—	$\mid \operatorname{dog}) = 2/3$	$\mid \operatorname{dog}) = 2/3$
pdf-dc99cf88_F00228	012	—	$\hat{\rho}_{\text{dog}} = \hat{\rho}_{\text{white dog}} + \hat{\rho}_{\text{black dog}}$	$\hat{\rho}_{\text{dog}} = \hat{\rho}_{\text{white dog}} + \hat{\rho}_{\text{black dog}}$
pdf-dc99cf88_F00229	013	—	$\hat{\rho}_{\text{dog}} = \hat{\rho}_{\text{small black dog}} + \hat{\rho}_{\text{big black dog}} + \hat{\rho}_{\text{small white dog}}$	$\hat{\rho}_{\text{dog}} = \hat{\rho}_{\text{small black dog}} + \hat{\rho}_{\text{big black dog}} + \hat{\rho}_{\text{small white dog}}$
pdf-dc99cf88_F00230	013	—	$V \otimes V$	$V \otimes V$
pdf-dc99cf88_F00231	013	—	$x_{i_3} x_{i_2} x_{i_1}$	$x_{i_3} x_{i_2} x_{i_1}$
pdf-dc99cf88_F00232	013	—	$x_{j_2} x_{j_1}$	$x_{j_2} x_{j_1}$
pdf-dc99cf88_F00233	013	—	$\hat{\rho}_{x_{i_3} x_{i_2} x_{i_1}}$	$\hat{\rho}_{x_{i_3} x_{i_2} x_{i_1}}$
pdf-dc99cf88_F00234	013	—	$\hat{\rho}_{x_{j_2} x_{j_1}}$	$\hat{\rho}_{x_{j_2} x_{j_1}}$
pdf-dc99cf88_F00235	013	—	$\hat{\rho}_{\text{tiny toy poodle}}$	$\hat{\rho}_{\text{tiny toy poodle}}$
pdf-dc99cf88_F00236	013	—	$\hat{\rho}_{\text{new park}}$	$\hat{\rho}_{\text{new park}}$
pdf-dc99cf88_F00237	013	—	$\hat{\rho}_{\text{I walked my tiny toy poodle at the new park}}$	$\hat{\rho}_{\text{I walked my tiny toy poodle at the new park}}$
pdf-dc99cf88_F00238	013	—	ρ'	ρ'
pdf-dc99cf88_F00239	013	—	$\rho \geq \rho'$	$\rho \geq \rho'$
pdf-dc99cf88_F00240	013	—	$\rho - \rho'$	$\rho - \rho'$
pdf-dc99cf88_F00241	013	—	$x_{i_{k+1}}$	$x_{i_{k+1}}$
pdf-dc99cf88_F00242	013	—	$\hat{\rho}_{\text{dog}} \geq \hat{\rho}_{\text{black dog}}$	$\hat{\rho}_{\text{dog}} \geq \hat{\rho}_{\text{black dog}}$
pdf-dc99cf88_F00243	013	—	$\hat{\rho}_{\text{dog}} \geq \hat{\rho}_{\text{white dog}}$	$\hat{\rho}_{\text{dog}} \geq \hat{\rho}_{\text{white dog}}$
pdf-dc99cf88_F00244	014	—	s'	s'
pdf-dc99cf88_F00245	014	—	$\rho_s \geq \pi(s' \mid s) \rho_{s'}$	$\rho_s \geq \pi(s' \mid s) \rho_{s'}$
pdf-dc99cf88_F00246	014	—	$\hat{\rho}_s \geq \hat{\rho}_{s'}$	$\hat{\rho}_s \geq \hat{\rho}_{s'}$
pdf-dc99cf88_F00247	014	—	$N \geq 1$	$N \geq 1$
pdf-dc99cf88_F00248	014	—	L	L
pdf-dc99cf88_F00249	014	—	$\sqcup_{k=1}^{N-1} X^k \times X$	$\sqcup_{k=1}^{N-1} X^k \times X$
pdf-dc99cf88_F00250	014	—	$s = (x_{i_k}, \dots, x_{i_1}, x_a)$	$s = (x_{i_k}, \dots, x_{i_1}, x_a)$
pdf-dc99cf88_F00251	014	—	$k \leq N-1$	$k \leq N-1$
pdf-dc99cf88_F00252	014	—	$x_{i_k} \cdots x_{i_1} x_a$	$x_{i_k} \cdots x_{i_1} x_a$
pdf-dc99cf88_F00253	014	—	$(x_{i_k}, \dots, x_{i_1}, x_a)$	$(x_{i_k}, \dots, x_{i_1}, x_a)$
pdf-dc99cf88_F00254	014	—	$s, s' \in L$	$s, s' \in L$
pdf-dc99cf88_F00255	014	—	$s \leq s'$	$s \leq s'$
pdf-dc99cf88_F00256	014	—	$dog \leq$	$dog \leq$
pdf-dc99cf88_F00257	014	—	$s \in L$	$s \in L$
pdf-dc99cf88_F00258	014	—	$s \leq s$	$s \leq s$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
pdf-dc99cf88_F00259	014	—	$s, s^{\prime \prime}, s^{\prime \prime \prime} \in L$	$s, s', s'' \in L$
pdf-dc99cf88_F00260	014	—	$s^{\prime \prime} \leq s^{\prime \prime \prime}$	$s' \leq s''$
pdf-dc99cf88_F00261	014	—	$s^{\prime \prime \prime}$	s''
pdf-dc99cf88_F00262	014	—	$s \leq s^{\prime \prime \prime}$	$s \leq s''$
pdf-dc99cf88_F00263	014	—	$X^N \cong X^{N-1} \times X$	$X^N \cong X^{N-1} \times X$
pdf-dc99cf88_F00264	014	—	$\pi: X^{N-1} \times X \rightarrow \mathbb{R}$	$\pi: X^{N-1} \times X \rightarrow \mathbb{R}$
pdf-dc99cf88_F00265	014	—	$s \in X^{N-1}$	$s \in X^{N-1}$
pdf-dc99cf88_F00266	014	—	$s = x_{i_k} \cdots x_{i_1}$	$s = x_{i_k} \cdots x_{i_1}$
pdf-dc99cf88_F00267	014	—	$\mathcal{L} \subseteq L$	$\mathcal{L} \subseteq L$
pdf-dc99cf88_F00268	014	—	$(\text{Pos}(V), \leq)$	$(\text{Pos}(V), \leq)$
pdf-dc99cf88_F00269	014	—	$(\mathcal{L}, \leq) \rightarrow (\text{Pos}(V), \leq)$	$(\mathcal{L}, \leq) \rightarrow (\text{Pos}(V), \leq)$
pdf-dc99cf88_F00270	015	—	$s^{\prime \prime} = x_{i_m} \cdots x_{i_{k+1}} s$	$s' = x_{i_m} \cdots x_{i_{k+1}} s$
pdf-dc99cf88_F00271	015	—	$1 \leq k \leq m \leq N-1$	$1 \leq k \leq m \leq N-1$
pdf-dc99cf88_F00272	015	—	$\hat{\rho}_{\text{black cat}} \leq \hat{\rho}_{\text{dog}}$	$\hat{\rho}_{\text{black cat}} \leq \hat{\rho}_{\text{dog}}$
pdf-dc99cf88_F00273	015	—	$\hat{\rho}_{\text{black cat}} = \frac{1}{5} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$	$\hat{\rho}_{\text{black cat}} = \frac{1}{5} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
pdf-dc99cf88_F00274	015	—	$\text{dog} \not\leq$	$\text{dog} \not\leq$
pdf-dc99cf88_F00275	015	—	$\mathcal{L}$	$\mathcal{L}$
pdf-dc99cf88_F00276	015	—	$\text{by}^3 \uparrow \{\text{dog}\} := \{s \in L \mid \text{dog} \leq s\}$	$\text{by}^3 \uparrow \{\text{dog}\} := \{s \in L \mid \text{dog} \leq s\}$
pdf-dc99cf88_F00277	015	—	$x_{i_k} \cdots x_{i_2} x_{i_1} \in \mathcal{L} \subseteq L$	$x_{i_k} \cdots x_{i_2} x_{i_1} \in \mathcal{L} \subseteq L$
pdf-dc99cf88_F00278	015	—	$x_{i_1} = \text{dog}$	$x_{i_1} = \text{dog}$
pdf-dc99cf88_F00279	015	—	$\uparrow \{\text{dog}\}$	$\uparrow \{\text{dog}\}$
pdf-dc99cf88_F00280	015	—	$\mathcal{L} \rightarrow \text{Pos}(V)$	$\mathcal{L} \rightarrow \text{Pos}(V)$
pdf-dc99cf88_F00281	015	—	$(P, \leq, \cdot, 1)$	$(P, \leq, \cdot, 1)$
pdf-dc99cf88_F00282	015	—	$(P, \leq)$	$(P, \leq)$
pdf-dc99cf88_F00283		—	$1 \in P$	$1 \in P$
pdf-dc99cf88_F00284		—	$L^{\text{op}} \rightarrow UL$	$L^{\text{op}} \rightarrow UL$
pdf-dc99cf88_F00285		—	UL	UL
pdf-dc99cf88_F00286		—	$\cdot: P \times P \rightarrow P$	$\cdot: P \times P \rightarrow P$
pdf-dc99cf88_F00287	016	—	$p q$	pq
pdf-dc99cf88_F00288	016	—	$p \cdot q$	$p \cdot q$
pdf-dc99cf88_F00289		—	$p, p^{\prime \prime}, q, q^{\prime \prime} \in P$	$p, p', q, q' \in P$
pdf-dc99cf88_F00290		—	$p \leq p^{\prime \prime}$	$p \leq p'$
pdf-dc99cf88_F00291		—	$q \leq q^{\prime \prime}$	$q \leq q'$
pdf-dc99cf88_F00292		—	$p q \leq p^{\prime \prime} q^{\prime \prime}$	$pq \leq p' q'$
pdf-dc99cf88_F00293		—	$1 p = p 1 = p$	$1p = p1 = p$
pdf-dc99cf88_F00294		—	$p \in P$	$p \in P$
pdf-dc99cf88_F00295		—	$(p q) r = p(q r)$	$(pq)r = p(qr)$
pdf-dc99cf88_F00296		—	$p, q, r \in P$	$p, q, r \in P$
pdf-dc99cf88_F00297		—	$p q = q p$	$pq = qp$
pdf-dc99cf88_F00298		—	$p, q \in P$	$p, q \in P$
pdf-dc99cf88_F00299	016	—	$[0, 1] \subseteq \mathbb{R}$	$[0, 1] \subseteq \mathbb{R}$
pdf-dc99cf88_F00300	016	—	$(\mathcal{V}, \leq, \cdot, 1)$	$(\mathcal{V}, \leq, \cdot, 1)$
pdf-dc99cf88_F00301	016	—	$\mathcal{V}$	$\mathcal{V}$
pdf-dc99cf88_F00302	016	—	$\mathcal{C}$	$\mathcal{C}$
pdf-dc99cf88_F00303		—	$\text{ob}(\mathcal{C})$	$\text{ob}(\mathcal{C})$
pdf-dc99cf88_F00304		—	$c, d, \dots$	$c, d, \dots$
pdf-dc99cf88_F00305		—	$\mathcal{C}(c, d) \in \mathcal{V}$	$\mathcal{C}(c, d) \in \mathcal{V}$
pdf-dc99cf88_F00306		—	c	c
pdf-dc99cf88_F00307		—	$1 \leq \mathcal{C}(c, c)$	$1 \leq \mathcal{C}(c, c)$
pdf-dc99cf88_F00308		—	$\mathcal{C}(c, d) \cdot \mathcal{C}(d, e) \leq \mathcal{C}(c, e)$	$\mathcal{C}(c, d) \cdot \mathcal{C}(d, e) \leq \mathcal{C}(c, e)$
pdf-dc99cf88_F00309		—	c, d, e	c, d, e
pdf-dc99cf88_F00310	016	—	$\mathcal{D}$	$\mathcal{D}$
pdf-dc99cf88_F00311	016	—	$f: \text{ob}(\mathcal{C}) \rightarrow \text{ob}(\mathcal{D})$	$f: \text{ob}(\mathcal{C}) \rightarrow \text{ob}(\mathcal{D})$
pdf-dc99cf88_F00312	016	—	$c, c^{\prime \prime}$	c, c'
pdf-dc99cf88_F00313	016	—	$\mathcal{L} \subseteq L \subseteq \sqcup_{k=1}^{N-1} X^k \times X$	$\mathcal{L} \subseteq L \subseteq \sqcup_{k=1}^{N-1} X^k \times X$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
pdf-dc99cf88_F00314	016	—	$s, s^{\prime} \in \mathcal{L}$	$s, s' \in \mathcal{L}$
pdf-dc99cf88_F00315	016	—	$1 \leq \mathcal{L}(s, s)$	$1 \leq \mathcal{L}(s, s)$
pdf-dc99cf88_F00316	016	—	$\mathcal{L}(s, s^{\prime}) \mathcal{L}(s^{\prime}, s^{\prime \prime}) \leq \mathcal{L}(s, s^{\prime \prime})$	$\mathcal{L}(s, s') \mathcal{L}(s', s'') \leq \mathcal{L}(s, s'')$
pdf-dc99cf88_F00317	016	—	$s, s^{\prime}$	s, s'
pdf-dc99cf88_F00318	016	—	$s \in \mathcal{L}$	$s \in \mathcal{L}$
pdf-dc99cf88_F00319	016	—	$\mathcal{D} \subseteq \text{Pos}(V)$	$\mathcal{D} \subseteq \text{Pos}(V)$
pdf-dc99cf88_F00320	016	—	$\hat{\rho}_s, \hat{\rho}_{s^{\prime}} \in \mathcal{D}$	$\hat{\rho}_s, \hat{\rho}_{s'} \in \mathcal{D}$
pdf-dc99cf88_F00321		—	$\pi(s^{\prime} \mid s)$	$\pi(s' \mid s)$
pdf-dc99cf88_F00322		—	$\pi(ts \mid s)$	$\pi(ts \mid s)$
pdf-dc99cf88_F00323		—	$s^{\prime}=t s$	$s' = ts$
pdf-dc99cf88_F00324		—	t	t
pdf-dc99cf88_F00325		—	$\text{dog} \mid \text{dog}$	$\text{dog} \mid \text{dog}$
pdf-dc99cf88_F00326		—	$\text{dog})=\pi(\text{black dog}) / \pi(\text{dog})$	$\text{dog}) = \pi(\text{black dog}) / \pi(\text{dog})$
pdf-dc99cf88_F00327	017	—	$1 \leq \mathcal{D}(\hat{\rho}_s, \hat{\rho}_s)$	$1 \leq \mathcal{D}(\hat{\rho}_s, \hat{\rho}_s)$
pdf-dc99cf88_F00328	017	—	$\mathcal{D}(\hat{\rho}_s, \hat{\rho}_{s^{\prime}}) \mathcal{D}(\hat{\rho}_{s^{\prime}}, \hat{\rho}_{s^{\prime \prime}}) \leq$	$\mathcal{D}(\hat{\rho}_s, \hat{\rho}_{s'}) \mathcal{D}(\hat{\rho}_{s'}, \hat{\rho}_{s'') \leq$
pdf-dc99cf88_F00329	017	—	$\mathcal{D}(\hat{\rho}_s, \hat{\rho}_{s^{\prime \prime}})$	$\mathcal{D}(\hat{\rho}_s, \hat{\rho}_{s''})$
pdf-dc99cf88_F00330	017	—	$\text{tr} \hat{\rho}_s=\pi(s)$	$\text{tr} \hat{\rho}_s = \pi(s)$
pdf-dc99cf88_F00331	017	—	$\text{tr} \hat{\rho}_{s^{\prime}} / \text{tr} \hat{\rho}_s=\pi\left(s^{\prime}\right) / \pi(s)=\pi\left(s^{\prime} \mid s\right)$	$\text{tr} \hat{\rho}_{s'} / \text{tr} \hat{\rho}_s = \pi(s') / \pi(s) = \pi(s' \mid s)$
pdf-dc99cf88_F00332	017	—	$\mathcal{L}(s, s^{\prime}) \leq \mathcal{D}(\hat{\rho}_s, \hat{\rho}_{s^{\prime}})$	$\mathcal{L}(s, s') \leq \mathcal{D}(\hat{\rho}_s, \hat{\rho}_{s'})$
pdf-dc99cf88_F00333	017	—	$\mathcal{L} \rightarrow \mathcal{D}$	$\mathcal{L} \rightarrow \mathcal{D}$
pdf-dc99cf88_F00334	017	—	$\mathcal{L}(s, s^{\prime})=\pi\left(s^{\prime} \mid s\right)$	$\mathcal{L}(s, s') = \pi(s' \mid s)$
pdf-dc99cf88_F00335	017	—	$\hat{\rho}_{s^{\prime}} \leq \hat{\rho}_s$	$\hat{\rho}_{s'} \leq \hat{\rho}_s$
pdf-dc99cf88_F00336	017	—	$\mathcal{D}^{\text {op }}\left(\hat{\rho}_{s^{\prime}}, \hat{\rho}_s\right):=$	$\mathcal{D}^{\text{op}}(\hat{\rho}_{s'}, \hat{\rho}_s) :=$
pdf-dc99cf88_F00337	017	—	$\mathcal{D}\left(\hat{\rho}_s, \hat{\rho}_{s^{\prime}}\right)=\text {tr} \hat{\rho}_{s^{\prime}} / \text {tr} \hat{\rho}_s=\pi\left(s^{\prime} \mid s\right)$	$\mathcal{D}(\hat{\rho}_s, \hat{\rho}_{s'}) = \text{tr} \hat{\rho}_{s'} / \text{tr} \hat{\rho}_s = \pi(s' \mid s)$
pdf-dc99cf88_F00338	017	—	$\left\langle\hat{\rho}_{s^{\prime}} x, x\right\rangle=\pi\left(s^{\prime} x\right) \leq \pi(s x)=$	$\langle\hat{\rho}_{s'} x, x\rangle = \pi(s' x) \leq \pi(s x) =$
pdf-dc99cf88_F00339	017	—	$\left\langle\hat{\rho}_s x, x\right\rangle$	$\langle\hat{\rho}_s x, x\rangle$
pdf-dc99cf88_F00340	017	—	$\pi\left(s^{\prime} x\right)>0$	$\pi(s' x) > 0$
pdf-dc99cf88_F00341	017	—	$\pi(s x)>0$	$\pi(s x) > 0$
pdf-dc99cf88_F00342	017	—	$x \mapsto-\log (x)$	$x \mapsto -\log(x)$
pdf-dc99cf88_F00343	017	—	$[0,1] \rightarrow[0, \infty]$	$[0, 1] \rightarrow [0, \infty]$
pdf-dc99cf88_F00344	018	—	$\psi_{\text {net }}$	ψ_{net}
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Unrecovered image regions — TikZ / table / failed-math candidates

Regions MathPix left as images (no LaTeX). Reconstruct with pdfdrill vision (LLM classify + transcribe to TikZ/tabular/math); verify an LLM result against the real ink with inkdrill.

Identifier	Page	Image	Source
pdf-dc99cf88_DIA_0001	008		tex.zip (local)
pdf-dc99cf88_DIA_0002	009		tex.zip (local)
pdf-dc99cf88_DIA_0003	013		tex.zip (local)
pdf-dc99cf88_DIA_0004	017		tex.zip (local)